

Overfitting and Underfitting of KNN

$k = \text{Low} \Rightarrow \text{Overfitting}$
 $k = \text{High} \Rightarrow \text{Underfitting}$

Limitations of KNN

Failure Case

- 1) Large dataset
- 2) High dimensional data
- 3) Outliers Prone
- 4) Non-homogeneous Feature scales
- 5) Imbalanced Dataset (Biased results)
- 6) If KNN is used for inference and not for predictions